Practice workbook · Proving shapes and straight lines with vectors

Using the fact that parallel vectors are scalar multiples of each other to prove lines parallel, points collinear or a quadrilateral a parallelogram or trapezium, or to find an unknown that makes two vectors parallel.

10 question parts: 1 worked-example retry and 9 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-VEC-06

1. Question 4 · part a · 3 marks

2020 JAN · Paper 02

Worked-example retry · Attempt part a only.

4 The points A, B, C and D are the vertices of a quadrilateral $ABCD$ such that

$\overrightarrow{AB} = 7\mathbf{i} + p\mathbf{j}$ $\overrightarrow{AC} = 11\mathbf{i} - p\mathbf{j}$ $\overrightarrow{AD} = 4\mathbf{i} - 2p\mathbf{j}$

(a) Show that, for all values of p, $ABCD$ is a parallelogram.

(3)

Given that $|\overrightarrow{BD}| = 3\sqrt{10}$

(b) find the possible values of p.

(3)

Given that $p > 0$

(c) find a unit vector which is parallel to $\overrightarrow{BD}$.

(1)

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1. Question 4 · part a · 3 marks · Working and answer

Working and answer

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2. Question 1 · part whole · 4 marks

2025 JUN · Paper 02 (regional R)

Further practice · Attempt part whole only.

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1 The points A, B, C and D are the vertices of a quadrilateral $ABCD$ such that

$$\overrightarrow{AB} = 3\mathbf{a} + 2\mathbf{b} \qquad \overrightarrow{AD} = 2\mathbf{a} - 5\mathbf{b} \qquad \overrightarrow{AC} = 5\mathbf{a} - 3\mathbf{b}$$

Show that $ABCD$ is a parallelogram.

(4)

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(Total for Question 1 is 4 marks)

2. Question 1 · part whole · 4 marks · Working and answer

Working and answer

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3. Question 10 · part a · 3 marks

2024 JUN · Paper 02

Further practice · Attempt part a only.

10 The points A, B, C and D are the vertices of a quadrilateral such that

$$\overrightarrow{AB} = 3\mathbf{a} + 4\mathbf{b} \quad \overrightarrow{AC} = 7\mathbf{a} + 9\mathbf{b} \quad \overrightarrow{AD} = 4\mathbf{a} + 5\mathbf{b}$$

(a) Show that $ABCD$ is a parallelogram.

(3)

BC is extended to the point E such that BCE is a straight line.

Point F lies on CD such that $CF : FD = 1 : 2$

Given that A, F and E are collinear,

(b) find the vector $\overrightarrow{AE}$ in the form $X\mathbf{a} + Y\mathbf{b}$ where X and Y are rational numbers to be found.

(8)

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3. Question 10 · part a · 3 marks · Working and answer

Working and answer

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4. Question 9 · part a · 4 marks

2025 NOV · Paper 02

Further practice · Attempt part a only.

9 The points A, B, C and D are the vertices of a quadrilateral where

$$\overrightarrow{AB} = 2\mathbf{a} + 4\mathbf{b} \qquad \overrightarrow{AC} = 7\mathbf{a} + 7\mathbf{b} \qquad \overrightarrow{AD} = 10\mathbf{a} + 6\mathbf{b}$$

(a) Show that $ABCD$ is a trapezium.

(4)

Given that $|\mathbf{a}| = 1$ and $|\mathbf{b}| = 1$ and that $\mathbf{a}$ and $\mathbf{b}$ are perpendicular to each other

(b) find a unit vector parallel to $\overrightarrow{BD}$

(4)

The point Y lies on CD such that $CY : YD = 1 : 2$

The lines AC and BY intersect at the point X

(c) (i) Use a vector method to find the ratio $BX : XY$

(4)

(ii) Hence find, in its simplest form, the ratio

area of triangle CXY : area of trapezium $ABCD$

(4)

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P 7 8 9 6 5 A 0 2 0 3 2

4. Question 9 · part a · 4 marks · Working and answer

Working and answer

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5. Question 10 · part b · 2 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part b only.

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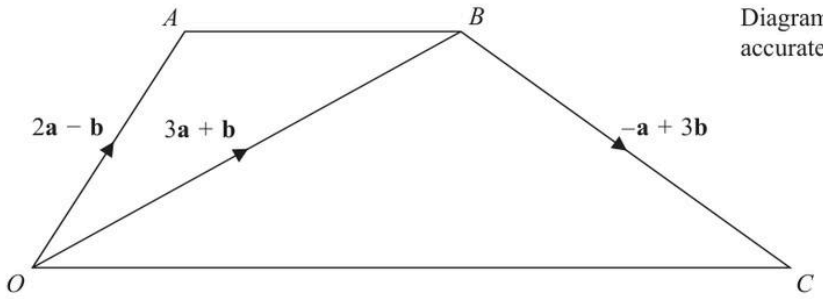


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows quadrilateral $OABC$ with

$$\overrightarrow{OA} = 2\mathbf{a} - \mathbf{b} \qquad \overrightarrow{OB} = 3\mathbf{a} + \mathbf{b} \qquad \overrightarrow{BC} = -\mathbf{a} + 3\mathbf{b}$$

(a) Find $\overrightarrow{AB}$ as a simplified expression in terms of **a** and **b**.

(2)

(b) Prove that $\overrightarrow{OC}$ is parallel to $\overrightarrow{AB}$

(2)

The diagonals, OB and AC, intersect at the point X.

(c) Using a vector method find the ratio $AX : XC$

(7)

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5. Question 10 · part b · 2 marks · Working and answer

Working and answer

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6. Question 9 · part a · 4 marks

2022 JUN · Paper 01 (regional R)

Further practice · Attempt part a only.

9

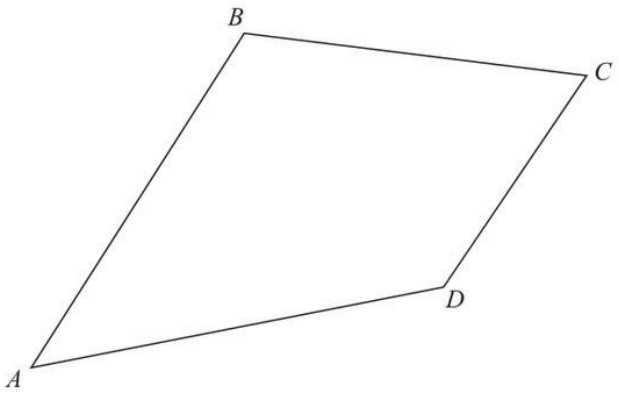


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Figure 3

Figure 3 shows quadrilateral $ABCD$ such that

$$\overrightarrow{AD} = 2\mathbf{a} + \mathbf{b} \qquad \overrightarrow{BC} = \frac{1}{3}\mathbf{b} \qquad \overrightarrow{BD} = -4\mathbf{a} - \mathbf{b}$$

(a) Prove that $\overrightarrow{AB}$ is parallel to $\overrightarrow{DC}$ **(4)**

The diagonals, AC and BD, of the quadrilateral intersect at the point Y.

(b) Using a vector method, find $\overrightarrow{AY}$ as a simplified expression in terms of **a** and **b** **(6)**

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6. Question 9 · part a · 4 marks · Working and answer

Working and answer

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7. Question 6 · part a · 4 marks

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Further practice · Attempt part a only.

6 The points P, Q, R and S are the vertices of a quadrilateral $PQRS$ such that

$$\overrightarrow{PQ} = 2\mathbf{i} + 3\mathbf{j} \qquad \overrightarrow{PR} = -\mathbf{i} + 18\mathbf{j} \qquad \overrightarrow{PS} = -3\mathbf{i} + 15\mathbf{j}$$

(a) Show that $PQRS$ is a parallelogram.

(4)

(b) Find a unit vector parallel to $\overrightarrow{QS}$ as a simplified expression in terms of **i** and **j**

(4)

The point T lies on QS such that $QT : TS = 5 : 8$

(c) Find $\overrightarrow{PT}$ as a simplified expression in terms of **i** and **j**

(2)

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7. Question 6 · part a · 4 marks · Working and answer

Working and answer

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8. Question 7 · part c · 4 marks

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

7 O, A, B and C are fixed points such that

$$\overrightarrow{OA} = 8\mathbf{i} - 6\mathbf{j} \qquad \overrightarrow{OB} = 15\mathbf{i} - 6\mathbf{j} \qquad \overrightarrow{OC} = 8\mathbf{i} + \mathbf{j}$$

(a) Find $\overrightarrow{BC}$ as a simplified expression in terms of $\mathbf{i}$ and $\mathbf{j}$

(2)

(b) Find a unit vector parallel to $\overrightarrow{BC}$

(2)

The point M is the midpoint of OA and the point N lies on OB such that $ON : NB = 1 : 2$

(c) Show that the points M, N and C are collinear.

(4)

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8. Question 7 · part c · 4 marks

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

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Question 7 continued

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8. Question 7 · part c · 4 marks

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Further practice · Attempt part c only.

Question 7 continued

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8. Question 7 · part c · 4 marks

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Further practice · Attempt part c only.



Question 7 continued

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(Total for Question 7 is 8 marks)

8. Question 7 · part c · 4 marks · Working and answer

Working and answer

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9. Question 11 · part b · 4 marks

2022 JAN · Paper 02 (regional R)

Further practice · Attempt part b only.

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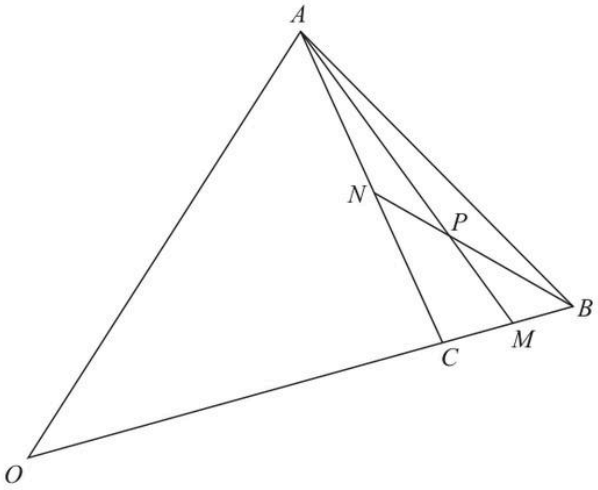


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Figure 3

Figure 3 shows triangle OAB with $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OB} = \mathbf{b}$

The point C lies on OB such that $OC : CB = 2 : 1$
The point M is the midpoint of CB and the point N is the midpoint of AC
The lines AM and NB intersect at the point P

(a) Using a vector method, find $\overrightarrow{OP}$ as a simplified expression in terms of **a** and **b**

(9)

The point Q is the midpoint of AB

(b) Using a vector method, show that C, P and Q are collinear.

(4)

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9. Question 11 · part b · 4 marks

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Further practice · Attempt part b only.

Question 11 continued

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9. Question 11 · part b · 4 marks

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Further practice · Attempt part b only.

Question 11 continued

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(Total for Question 11 is 13 marks)

9. Question 11 · part b · 4 marks · Working and answer

Working and answer

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10. Question 4 · part a · 6 marks

2023 JAN · Paper 01 (regional R)

Further practice · Attempt part a only.

4 O, A and B are fixed points such that

$$\overrightarrow{OA} = p\mathbf{i} + 2p\mathbf{j} \qquad \overrightarrow{OB} = 5\mathbf{i} + 9p\mathbf{j}$$

Given that $\overrightarrow{AB}$ is parallel to $(\mathbf{i} - 2\mathbf{j})$

(a) find the value of p

(6)

(b) Hence find $\overrightarrow{AB}$ as a simplified expression in terms of $\mathbf{i}$ and $\mathbf{j}$

(2)

(c) Find a unit vector parallel to $\overrightarrow{OA}$

Give your answer in the form $\frac{\sqrt{a}}{5}(b\mathbf{i} + c\mathbf{j})$ where a, b and c are integers to be found.

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P 7 1 8 1 8 A 0 8 2 8

10. Question 4 · part a · 6 marks · Working and answer

Working and answer

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